SplitSmart

Scan a receipt, give each item to the people who actually shared it, and settle up with the fewest payments. One HTML file — no account, no backend, no build step. Nothing leaves the device unless you share a group.

Live
nothing4601.github.io/splitsmart

Code
github.com/Nothing4601/splitsmart

Tests
38 passing · node --test

Tier 1

CORE · ALL 7

- ✓ **Groups & members** — create, rename, delete; add and remove people; switch from one dashboard
- ✓ **Capture** — upload or camera, with a preview before reading
- ✓ **OCR & parsing** — Tesseract in a worker with a progress bar; items, quantities, prices, CGST/SGST, service, discount and total; addresses, GSTIN and footers dropped
- ✓ **Review** — every field editable, lines added or deleted, low-confidence words highlighted, photo beside the table
- ✓ **Validation** — shows the exact difference when the lines don't reach the printed total
- ✓ **Manual expense** — description, amount, payer, participants, date
- ✓ **Balances** — each member's net, recomputed on every change

Tier 2

ADVANCED · ALL 7

- ✓ **Itemized assignment** — tap people onto a line; shared items split between them; unassigned lines flagged before saving
- ✓ **Four split modes** — equal, exact, percent, shares, with the input validated
- ✓ **Proportional tax** — tax, service and discounts follow each person's item subtotal, not a head count
- ✓ **Rounding** — integer paise throughout; largest-remainder allocation, so shares always sum to the bill
- ✓ **Debt simplification** — exact minimum payments by dynamic programming; mark any settled
- ✓ **Categories** — zero-shot model, loaded on first use; your override always wins
- ✓ **Persistence** — localStorage for the data, IndexedDB for receipt photos

Tier 3

BROWNIE POINTS · ALL 9

- ✓ **UPI settle-up** — deep link and QR per payment
- ✓ **Analytics** — by category, member, group and month
- ✓ **Share, no backend** — JSON export/import, or a link
- ✓ **PWA & offline** — installable; every byte ships with the site
- ✓ **Recurring** — monthly items add themselves
- ✓ **Export** — PDF summary, CSV expenses
- ✓ **Batch scan** — a queue of receipts in one go
- ✓ **Accessibility** — keyboard, ARIA, focus rings, reduced motion
- ✓ **Live deployment** — GitHub Pages, public URL

Beyond the brief

Voice entry. Speak an expense — “Dev paid 600 for pizza, split equally” — or say who had what on a scanned bill: “I had the vada, the rest between the others.” Whisper runs on the device.

Live sync. Share a group by link and two phones stay in step, AES-GCM encrypted through a public relay. No account, no server of ours.

Cross-group net position. Everything you owe and are owed, netted across all groups and matched by person.

3D character avatars with a mix-and-match creator, and a pointer layer that lights up both people in a settlement row.

How it is built

One file. All of the app is `index.html` — CSS, markup and script inline. No bundler, no framework, no backend. three.js is vendored, so neither the first paint nor offline use depends on a CDN.

Money is never a float. Integer paise from parsing to settle-up. A documented largest-remainder rule hands out the odd paisa, so ₹100 three ways is 33.34 + 33.33 + 33.33 and the parts always rebuild the whole.

Tested, not trusted. A DOM-free `split-logic` block holds allocation, shares, balances, settle-up and the parsers. 38 unit tests pull it out of the HTML and run it in a sandbox.

Nothing heavy on first paint. OCR, the category model and Whisper load only when first used.